☆ Set theory for OHE

- Set is a finite well defined collection of distant elements.

$$A = \{1, 2, 3, 4\}$$

This is element of set.

- Cardinality of a set is no. of distant element present in set.

In this case $|A| = 4$

- Membership of set indicates weather an elements belongs to a set.

If it is there then true.
If it is not then False

for example lets take a set of a fuel

F = {Petrol, Disel, CNG}

Now If we want check weather Petrol is there

$$I_{petrol}(x) = \begin{cases} 1 & \text{if } x = \text{petrol} \\ 0 & \text{otherwise} \end{cases}$$

the output produce is

[1, 0, 0]

for disel output would be

[0, 1, 0]

for CNG Output would be

[0, 0, 1]

Puting that in a table will give one hot encoded value

fuel	Petro_f	Disel_f	CNG_f
Petrol	1	0	0
Disel	0	1	0
CNG	0	0	1

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- Set union of set (U):

The union of 2 sets contain all unique value in either set

for example

$X = \{1, 2, 3\}$

$Y = \{1, 2, 4, 5\}$

$X \cup Y = \{1, 2, 3, 4, 5\}$

Imagine a train dataset contains $A = \{1, 2, 3\}$ and testing dataset has $B = \{2, 3\}$

The combined categories are

$A \cup B = \{1, 2, 3\}$

Now here Feature '3' is un known for training data and and might cause problem.

To solve that we use Ohe =

Ohe = One Hot Encoder (handle_unknown = 'ignore')

which ignores unknown column from testing

2. Intersection of sets (∩):

The intersection of a set contains only element common to both sets.

A = {1, 2}

B = {2, 3, 4}

A ∩ B = {2}

Show for example a training data ⇒ A = {1, 2, 3}
testing data ⇒ B = {1, 2, 3, 4}

A ∩ B = {1, 2, 3}

The 4 cap is not present in training data so it model doesn't learn from it.

handle-unknown = "ignore" functions like this

3. Difference of set: It is difference of 2 sets contains the element that belongs in first set but not on second set.

Symbol: (− or /) For example

A = {1, 2, 3} B = {3, 2, 4, 5}

A = {1} A − B = {1}, B − A = {4, 5}

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